

Database Administration

Lab: NoSQL and MongoDB Primer.

Andrés Oswaldo Calderón Romero, Ph.D.

April 27, 2026

1 Introduction

In this lab, we will continue our exploration of NoSQL technologies by focusing on MongoDB, one of the most widely used NoSQL databases. Through selected readings and hands-on labs, you will gain a foundational understanding of NoSQL types, use cases, and essential MongoDB operations. Although no formal submission is required, developing familiarity with MongoDB is crucial for your upcoming project work. This lab is designed to help you grasp basic concepts and commands that will be vital for your success in group activities and real-world applications.

2 Reading Material

To complement what we have covered about NoSQL in class, we will explore additional readings from the Coursera MOOC “*Introduction to NoSQL Databases*” by IBM and [Skills Network](#), available [here](#). The materials are:

1. NoSQL Types and Use Cases ([en](#), [sp](#)).
2. NoSQL Database Deployment Operations ([en](#), [sp](#)).
3. NoSQL Glossary ([en](#)).

3 Introduction to MongoDB

You will explore the basis of MongoDB through some introductory videos provided by Skills Network. You can get access to the following videos in the corresponding links:

1. MongoDB: [Overview](#).
2. MongoDB: [Advantages](#).
3. MongoDB: [Use Cases](#).

(*Optional*) We will give a try on the trial mode of Coursera and focus on the second module of the course (available [here](#)). For this activity, you must log in to Coursera and register for the course in order to gain access ([link to Module 2](#)). You will need follow the requirements to get the trial mode for this. Remember that this request is completely optional, you can skip it if you want.

In the module, we will concentrate on the Labs, particularly:

- Lab: Getting Started with MongoDB (Basics of MongoDB section).
- Lab: MongoDB CRUD (Getting Started with MongoDB section).
- Lab: MongoDB Indexing (Getting Started with MongoDB section).

4 Fundamental Concepts in MongoDB

The official documentation serves as the primary authority for exploring key MongoDB topics. Additionally, MongoDB offers a robust educational platform known as [MongoDB University](#), which provides a comprehensive catalog of essential resources for users ranging from beginners to advanced practitioners, including both free and paid content. For the scope of the final project, we recommend reviewing the following sections:

- Relational to Document Model:
<https://learn.mongodb.com/courses/relational-to-document-model>
- CRUD Operations in MongoDB:
<https://learn.mongodb.com/courses/crud-operations-in-mongodb>
- Fundamentals of Data Transformation:
<https://learn.mongodb.com/courses/fundamentals-of-data-transformation>
- Indexing Design Fundamentals:
<https://learn.mongodb.com/courses/indexing-design-fundamentals>
- MongoDB Indexes:
<https://learn.mongodb.com/courses/mongodb-indexes>
- MongoDB Indexes II:
<https://learn.mongodb.com/courses/mongodb-indexes-ii>

5 What We Expect

Actually, you do not have to submit any evidence of your work, but you will need to become familiar with MongoDB to work on the project. It is important for your group that you grasp the basic concepts of its use and fundamental commands.

Happy Hacking! 😎